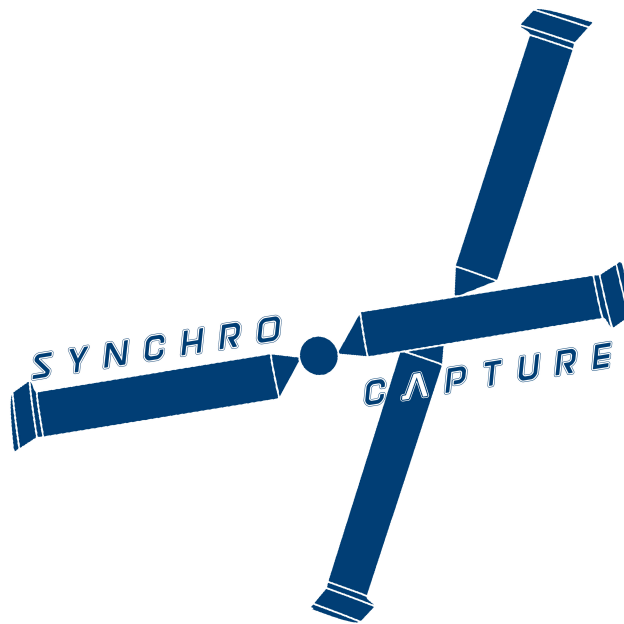


AERO96005 - GROUP DESIGN PROJECT
WHOLLY REUSABLE LAUNCH SYSTEM (RECOVERY)

Capture Architecture Design for the Complete Aerial-Recovery of an Orbital Launch System



Student: Samuel Jenner
CID: 01384152

Project Supervisor: Dr. E. Levis
Team Supervisor: Dr. M. Ribera Vicent
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Department of Aeronautics
South Kensington Campus
Imperial College London
London SW7 2AZ

Abstract

The preliminary design of the capture architecture for the aerial recovery of an orbital launch vehicle is presented. **3rd Generation Mid-Air Retrieval** is implemented and augmented to improve the safety and reliability of recapture, most notably the recovery vehicle in **remotely piloted**. A novel grapple control device is proposed being **1m in chord** and **1.08m in span**. It is controlled by a **Fishbone Active Camber** mechanism consisting of a **ABS plastic spine**, **Elastometric Matrix Composite skin** and **tendon pulley**. The grapple line is **7.62m** long and manufactured from **11mm thick Dyneema**. A **2 stage force model** is proposed to simulate the load history experienced during recovery. Furthermore, the preconceptual sizing of candidate vehicle concepts is presented.

WRR Team Members

Table 1: Team Members and Roles.

Name	Team Name	Team Code
Alessandro Bullitta Yassine Ghenima Cyril Gliner	Project co-ordination	WRR01
Chi Wing Cheng Alessandro Rossi Polvara Tin Hang Un Le Yin	Aerodynamics	WRR02
Anton Le Tejasva Malhotra Atri Sharma Alastair Wood	Structures & CAD	WRR03
Omar Arangath Thomas Cross Zuzanna Janusz Sam Jenner Jo Tan Tom Walker	Systems & Propulsion	WRR04
Filip Bastien Ananya Dubey Kai Hon Andrew Ng India Tavares	Flight Mechanics, Performance & Control	WRR05
Tom Alston Hakan Serpen Tze Leung Wong	Flight Simulation & Testing	WRR06

†Sub-team leads are shown in **bold**

Contents

1	Introduction	1
2	Preconceptual Weight and Power Sizing of Tiltrotor Concepts	1
2.1	Empty Weight Estimation	1
2.2	Power and Fuel Mass Estimation	1
3	Selection of Recapture Concept	1
3.1	Overview of Mid-Air Retrieval	2
3.1.1	First Generation MAR	2
3.1.2	Second Generation	2
3.1.3	Third Generation	2
3.2	Architecture Selection	3
3.3	Recapture Mission Profile	4
4	Grapple Hook Design	5
4.1	Automatic Closure Mechanism	5
4.2	Closure Action	5
4.3	Material Selection	5
4.4	Sizing	5
5	Grapple Line Design	6
5.1	Length	6
5.2	Material and Diameter Selection	6
6	Payload Equilibrium During Return Cruise	6
6.1	Methodology	6
6.2	Results	8
7	Grapple Control Device	8
7.1	Configuration	8
7.2	Control Surface Design	8
7.3	Control Surface Actuation	9
7.4	Sizing	10
8	Simulation of Recapture Force	10
9	Conclusion	12

References	13
A Preliminary Weight and Power Sizing of Manned and Unmanned Tiltrotor Concepts	15
B Payload Equilibrium Condition Results	15
C Grapple Rudder Schematic	16
D Detailed Schematics of the Recovery Procedure	17

List of Tables

1	Team Members and Roles.	ii
2	Summary of the cable lengths as indicated in Figure 3 for the first and second stage. . . .	4
3	Summary of the mass of each of the grapple components.	6
4	Summary of the drag coefficients for the components of the slung line.	7
5	Summary of the results of the payload equilibrium during return cruise.	8
6	Summary of the mass components of the optimal rudder.	10
7	Summary of the initial sizing of a manned and unmanned tiltrotor concept.	15

List of Figures

1	Evolution between 1GMAR and 2GMAR architecture.	3
2	Current 3GMAR architecture.	3
3	Schematic of the entire slung line after capture.	4
4	Schematic of the idealised slung line and the algorithm used to calculate the tensions, angles and diameter.	7
5	Schematic of the FBAC control surface actuation mechanism.	9
6	Selection of the optimal surface area for maximum control authority and the resulting range of permissible longitudinal and vertical positions.	11
7	Flight test data of a recovery attempt and the resulting load history.	12
8	Schematic of the equilibrium state of the payload during return cruise.	15
9	First stage equilibrium position convergence results.	16
10	Second stage equilibrium position convergence results.	16
11	Side view schematic of the grapple and grapple rudder.	17
12	Recapture step 1.	18
13	Recapture step 2.	18
14	Recapture step 3.	19
15	Recapture step 4.	19
16	Recapture step 5.	20

1 Introduction

The capability of an orbital launch system to be wholly reusable presents significant cost savings and drastically reduced turnaround times. This Group Design Project aims to present the preliminary design of such a system that has the capability to aerially recover a falling rocket stage. This report outlines the work done to design the architecture required for aerial recapture along with a detailed description of the recapture mission profile, additionally the preconceptual sizing of candidate vehicle concepts is presented.

The design of such a system is highly interdisciplinary, with system selection greatly affecting all aspects of the vehicle; from weight of systems onboard the launch vehicle to aerodynamic and structural considerations the recapture procedure places on the selected recovery vehicle. Along with the system design these trade-offs and their effects on each design discipline are explored to ensure a reliable and realistic recovery system architecture is produced.

2 Preconceptual Weight and Power Sizing of Tiltrotor Concepts

This section provides a brief overview of the work done to estimate the weight and power requirement of both a manned and unmanned tiltrotor concept which formed part of the vehicle selection process. Details of the results can be found in §A.

2.1 Empty Weight Estimation

Given the dependence on the aircraft to carry a large payload, a regression analysis between payload weight capacity and empty weight was used to determine the empty weight of the manned tiltrotor. No extrapolation was required, however, as a result of the niche configuration, the correlation of the historical data was poor. Despite this it was deemed acceptable to use as it served only as an order of magnitude comparison to the other configurations.

When sizing the unmanned tiltrotor, historical data could not be used as the current payload capacity of unmanned tiltrotors is several orders of magnitude lower than that required for the mission and hence large extrapolations would be needed. Instead, a percentage reduction in the empty weight of the manned configuration was defined and is outlined in [1].

2.2 Power and Fuel Mass Estimation

Given the estimated empty weight, the power requirement for each configuration was estimated by analysing the power requirement in the most restrictive case. This was done using actuator disk theory and also accounting for the power lost in turboshaft engines due to altitude.

Once the power requirement was found, fuel masses for cruise and hover were estimated using the fuel consumption of the Pratt and Whitney PT6C-67A, a turboshaft engine with a comparable power output. It is noted that the power consumption during cruise can be estimated as 60% of the power requirement at takeoff [2].

3 Selection of Recapture Concept

One of the critical design drivers in the vehicle selection process was the feasibility of the recapture concept. Therefore, along with the preconceptual design of the helicopter, fixed wing and VTOL concepts it was decided to outline several broad concepts for each of these designs for the recapture system. Once the vehicle was chosen, and being later in the project, more information from the launch team was available, the most appropriate system was selected to begin the conceptual and preliminary design.

Many exotic ideas were considered; a detailed outline of these and their advantages and disadvantages can be found in [3] and [4]. However, it quickly became apparent that a “hook and parafoil” concept, commonly referred to as Mid-Air Retrieval (MAR) in the literature, would be the most effective. This section outlines the concept of MAR, the reasons for selection and a detailed mission profile of the recapture process along with an overview of the system architecture.

3.1 Overview of Mid-Air Retrieval

MAR is by no means a new technology, and for the last 60 years has been used to recover payloads ranging from satellites to skydivers. The development of MAR systems is commonly split into 3 categories: 1st, 2nd and 3rd Generation (1G, 2G and 3G). This section aims to present the reader with an overview of the development of MAR systems along with their advantages and disadvantages in order to provide some context in the decision making process of the system architecture.

3.1.1 First Generation MAR

In the 1960s, developed by the U.S. Air Force in conjunction with the Central Intelligence Agency, the Corona programme aimed to inform the U.S. of Soviet weapons systems developments during the cold war by use of low resolution photo surveillance satellites [5]. In order to prevent the photographic film from disintegrating in the ocean when returning from orbit, 1GMAR was used to recover the film that was ejected from the Discoverer 14 satellites before impact. Engagement hooks were connected to poles that trailed behind a C-119 in order to intercept a round parachute, see Figure 1a. The hooks were linked to a constant tension winch designed to payout slack at around 1.25 times the weight of the payload to reduce any transient effect on the aircraft [6]. The specialised winches were heavy and mechanically complex and the small, unstable parachute was difficult for the pilot to track. Very precise control was required to intercept the payload and high relative velocities caused significant transience despite the winch. Payloads up to 1300 kg were recaptured, proof of concept designs were made for 2300 kg payloads but never tested.

3.1.2 Second Generation

2GMAR involved upgrading the existing round parachute to a more stable and controllable ram air inflated parafoil. Flying along a straight path at reduced velocities allowed easier tracking of the payload and lower relative velocities during recapture reduced transient effect on the aircraft’s dynamics. Initial 2GMAR configurations were rather complex, some involving seven parachutes to decelerate and recapture payloads [7]. However, the technology was improved and the Genesis Sample Return Capsule was recovered in 2004 using only one parachute, see Figure 1b. Despite this there were significant issues involved in the towing of the payload. Above certain velocities the parafoil would re-inflate causing significant stability issues.

3.1.3 Third Generation

By trailing a drogue line behind the parafoil that is in direct contact with the payload and the addition of a slider reefing system and an aerodynamically controlled grapple hook, much heavier loads can be captured [6]. A slider plate with reinforced grommets in each corner is fed through the parafoil suspension lines and directly connected to the drogue line. When the drogue line is pulled up by the aircraft the slider moves up the suspensions lines and, acting like a funnel for the parafoil, deflates the cells by drawing the ends of the parafoil together. This approach is significantly more effective, with close to zero relative velocities during recapture, it is demonstrated to be capable of capturing up to 80% of the payload capacity of an aircraft and current technologies indicate the potential to recover payloads of up to 10 tonnes, well above that required for this mission [6]. United Launch Alliance demonstrated the effectiveness of 3GMAR in 2008 capturing a skydiver, see Figure 2a. More recently, PDG Aviation Services and Rocket Lab have



(a) 1GMAR configuration on the C-119J in the Corona missions [5].



(b) 2GMAR configuration used in the recovery of Genesis Sample Return Capsule [8]

Figure 1: Evolution between 1GMAR and 2GMAR architecture.

demonstrated the use of 3GMAR to perform the recapture of falling rocket stages, see Figure 2b.



(a) United Launch Alliance demonstration of the recapture of a skydiver using 3GMAR [6].



(b) Rocket Lab demonstration of the recapture of a rocket stage [9].

Figure 2: Current 3GMAR architecture.

3.2 Architecture Selection

It is clear from the recent demonstrations of this technology that 3GMAR is the optimal Recovery procedure compared to the exotic solutions considered and is the selected method for this project.

The parafoil ensures a steady, slower glide slope and weighs less than a conventional parachute allowing for a lighter launch vehicle and easy tracking of the payload. The slider reefing system can be made from a lightweight synthetic material having negligible mass compared to a constant tension winch while being less mechanically complex. By evenly distributing the transfer of load from the parafoil to the aircraft over greater time scales, it has been shown that load factors of only 1.2 are expected during recapture [10]. This results in a lighter air frame that does not have to take excessive recapture loads. Furthermore, the aerodynamically controlled grapple reduces any swinging of the grapple line and can be used to fine tune its position. For these reasons 3GMAR was selected, however, novel solutions are proposed to augment the existing architecture to further increase the reliability and safety of this method. Most notably the helicopter will be remotely piloted drastically reducing the safety of the mission.

The system architecture is summarised in Figure 3 and the relevant lengths detailed in Table 2. Note the design of the parafoil is completed in [21], however not without strict restrictions imposed by the requirements of 3GMAR. Some of the design considerations were that the horizontal velocity must be

greater than 1.2 times the aircraft translational speed, which for a helicopter of this configuration will result in a minimum speed of around 25 knots [6]. Furthermore, the glide slope must be shallow enough such that the helicopter does not enter vortex ring state while tracking the payload. By considering these factors and the mass penalty of shallower, slower glide slopes a glide angle of 15 degrees was selected.

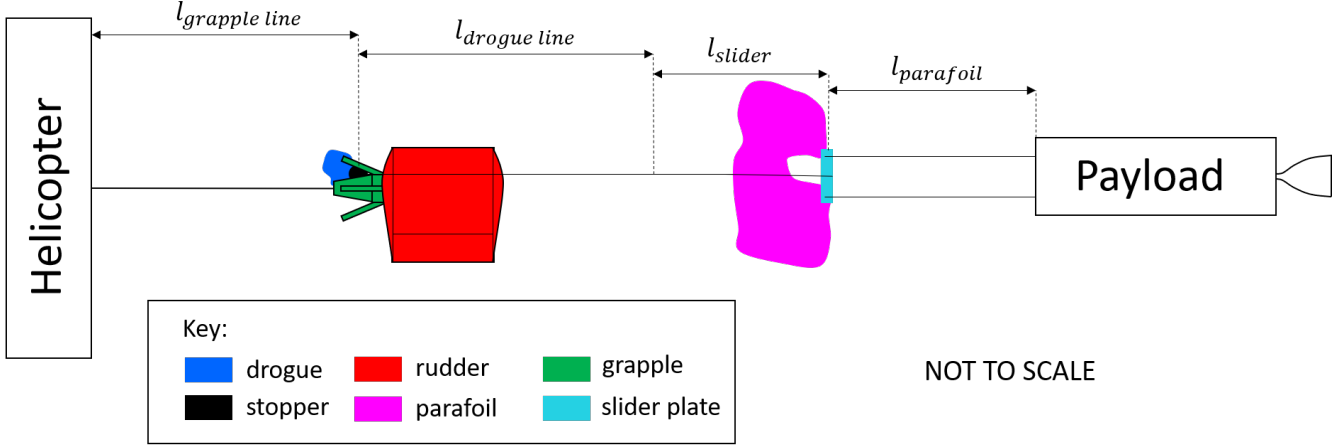


Figure 3: Schematic of the entire slung line after capture.

Table 2: Summary of the cable lengths as indicated in Figure 3 for the first and second stage.

Stage	$l_{grapple\ line}$	$l_{drogue\ line}$	l_{slider}	$l_{parafoil}$
First	7.62	14.9	7.45	14.9
Second	7.62	11.6	5.8	11.6

3.3 Recapture Mission Profile

Following from the selection of MAR as the recovery method, more specifically an augmented version of 3GMAR, the precise mission profile was formulated. This mission profile considers the segment after which the payload is within visual range of the aircraft as the entire recapture procedure is conducted by eyesight, through the on board cameras. For detailed schematics of each step, the reader is encouraged to follow Figures 12 through 16 and for information regarding acceptable limits for the completion of the manoeuvre please refer to [30]. The procedure can be broken down into 5 clear steps:

1. Upon confirmation of the aircraft being within visual range, the pilot follows the trajectory of the payload in an echelon formation. The design of the grapple hook means that either a right or left echelon formation is possible. The aim is to have the vertical and longitudinal position of the midpoint of the grapple line such that it intersects the midpoint of the drogue line, with a predefined lateral offset, in order to increase any margin of error in tracking the payload trajectory, maximising the chances of recapture.
2. The aircraft moves laterally such that the drogue line and the grapple meet.
3. Following confirmation of the connection of the drogue and grapple line, the aircraft begins to climb out of the parafoil glide slope until the drogue line is caught by one or more of the grapple tines. This triggers the a sensor on the grapple which closes the tines trapping the drogue line.
4. Following visual confirmation that the drogue line is secured in the closed grapple tines the aircraft continues climbing, this moves the grapple further up the drogue line until a stopper underneath the drogue is trapped by the grapple hook.

5. The aircraft continues climbing, pulling all the lines into tension. As a result, the slider panel is pulled up the parafoil suspension lines and the parafoil is reefed thus transferring the payload loads onto the aircraft. The recovery is complete.

This concludes the conceptual design of the recapture architecture. The following sections detail the preliminary design of the individual systems.

4 Grapple Hook Design

The grapple hook acts as the physical contact point between the payload and the aircraft, holding the stopper behind one or more of its tines, it has to be remotely actuated and a novel solution involving infrared (IR) signals, which does not require the actuation by the pilot, is proposed.

4.1 Automatic Closure Mechanism

For the grapple to close automatically upon the drogue line settling in one or more of the grapple's tines, an IR emitter and receiver will be implanted into the grapple body and each of the grapples tines respectively. The drogue line will disrupt the IR signal causing the tines to rapidly close. The emitter and receiver are both very small and have negligible mass compared to the grapple structure meaning they can be implanted without affecting the structural integrity.

4.2 Closure Action

The grapple will open and close much like the mechanism on a winged corkscrew, using a rack and pinion configuration. The rack will be driven by an electrical linear actuator powered by a 12V lithium ion battery. This saves on mass, complexity and volume of an equivalent hydraulic actuator [11]. Redundancies will be provided by an additional two power sources and another actuator, the high level of redundancy is required because if the grapple does not close then the payload cannot be recovered. The batteries will also power the IR emitter and receiver. A small camera mounted above the grapple will be used for visual confirmation of the drogue line being held in the grapple tines.

4.3 Material Selection

The grapple itself will be manufactured from grade 5 titanium. Other materials considered included stainless steel and carbon fibre reinforced polymers (CFRPs). The material selection was driven by strength to weight ratio, impact resistance and corrosion resistance. Operating in a maritime environment there is a high risk of salt water corrosion and grade 5 titanium is particularly resistant to this, increasing the service life and reducing maintenance costs compared to stainless steel [12]. Furthermore, the repeated impact loading of the tines upon closure will pose risks if CFRPs are implemented, hence grade 5 titanium was selected.

4.4 Sizing

Sizing of the grapple hook was done by using a typical grapple for slung line operations, rated to the required load, and adjusting the mass to account for titanium being used [17]. It is noted that the sizing is somewhat arbitrary and further design iterations must consider the effect of varying the dimensions of the grapple on the reliability of recapture. The sizing of the actuators and power supply was done so using [11]. The volumes, masses and actuator characteristics were confirmed by comparing the results to current state of the art systems and they were found to be equivalent [13], [14]. The results of the sizing is presented in Table 3.

Table 3: Summary of the mass of each of the grapple components.

Component	Grapple structure	Actuators	Power supply	Camera
Mass (kg)	5	1.5	0.73	0.15

5 Grapple Line Design

In order to reduce large moments on the airframe during return cruise and recapture the structures team was collaborated with to ensure as low a center of gravity as possible [15]. The length of the line affects the stability of the helicopter during the return cruise which is discussed in §6.

5.1 Length

Given the dependence on the validation of stability models and structural constraints to confirm that a particular length of grapple line is acceptable, information about the lengths of previous 3GMAR grapple lines were used as a starting point for the design. The initial length selected was 7.62m based on 3GMAR test that indicated the procedure with that line length was acceptable [6].

5.2 Material and Diameter Selection

The design drivers for the material selection were abrasion and corrosion resistance, strength to weight ratio and fatigue resistance from multiple repeated loading cycles. It was clear that Dyneema SK78 was the obvious choice, being far tougher, stronger and lighter than any alternative natural or synthetic fibres [16]. It is also commonly used in helicopter slung line applications [17].

Following the analysis in §6, the minimum possible diameter for the grapple line was found. However, to reduce the cost of having custom made long lines and, to reduce delays if a line needed replacing, it was decided to use the next closest diameter that is commonly available for purchase. As a result the diameter of the slung line was found to be 11mm (7/16" from [17]).

6 Payload Equilibrium During Return Cruise

Following the selection of the initial length of the grapple line and confirmation of the lengths of the drogue line, parafoil suspension lines and slider length from the launch team, the return cruise mission segment was analysed by the flight mechanics team to assess the stability of the slung load. In order to perform the analysis, the equilibrium position of the slung line must be calculated. The methodology and results of this calculation, for both the first and second stage, are presented here.

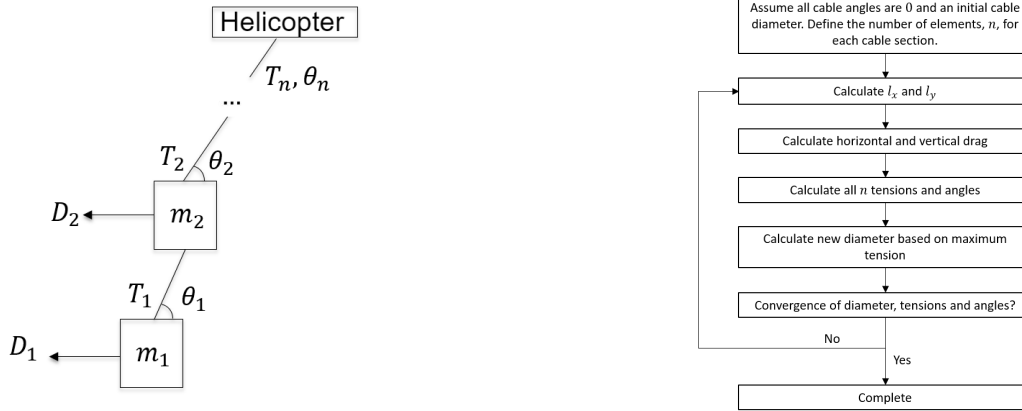
6.1 Methodology

So that the distribution of mass and drag along the slungline can be accounted for a very simple numerical method was devised in order to calculate the equilibrium position. The slungline was modelled as a series of point masses connected by massless, inextensible cables. Each point mass and cable represents a single element, there are n elements in the entire slung line. The systems suspended from the slungline can also be modelled in this way by adding additional point mass elements to the cable. Figure 10a shows a schematic of the idealised systems, note it only displays horizontal drag. Using equations of equilibrium in 2 directions at each element, $2n$ equations are simultaneously solved for $2n$ unknowns.

In order to account for the inclination of the slung line creating an elliptical cross section in the freestream, and the effect of any decent velocity of the helicopter, the calculation and iteration of effective lengths was implemented. Calculating the horizontal drag components was done so by taking the vertical component

of the cable length as the associated length scale and the horizontal length component for the vertical drag. The advantage of this simplification is that no further analysis needed to have been completed to evaluate the variation of the drag coefficient of the cable with its inclination which would be unnecessarily accurate given the assumptions already made. After each iteration the horizontal and vertical components of the cable length, l_x and l_y , are recalculated until convergence. The effect of the downwash from the blades is neglected as it was deemed to have negligible impact.

An additional iteration that must be completed is the diameter of the grapple line. The thicker the line the greater the force it can take, however the larger the drag. After each iteration the maximum tension in the grapple line was used to re-evaluate the diameter required after applying an ultimate load factor of 7, common practice for non-human external cargo lines [17]. Figure 10b outlines the algorithm used to calculate the tensions and angles in the slung line.



(a) Schematic of the idealised slung line. Note for clarity only the horizontal drag is shown.

(b) Algorithm for the idealised slung line system.

Figure 4: Schematic of the idealised slung line and the algorithm used to calculate the tensions, angles and diameter.

In order to calculate the drag on each element, the drag coefficient of the cable and the various systems on the slung line were evaluated and the results are summarised in Table 4. It is noted that during the implementation of the method, changes of the cable diameter are assumed to not affect the drag coefficient of the cable, this is reasonable given that the tension in the cable is dominated by the drag and mass of the payload. For further details of the evaluation of the reefed parafoil, first and second stage drag coefficients please refer to [18]. The drag contribution of the drogue, and grapple rudder are assumed negligible and the grapple itself is modelled as a cylinder using [20]. Cables with the same outer coating and lay were used to estimate the contribution of the cable [19]. For a breakdown of the diameters, lengths and quantity of the drogue and parafoil suspension lines please refer to [21].

Table 4: Summary of the drag coefficients for the components of the slung line.

Component	Cable	Grapple	Reefed parafoil	First stage	Second stage
C_d	1.02	1.00	0.97	0.82	0.77

One issue that arose in the drag breakdown was finding the reference area for the reefed parafoil. Being a complicated shape and the fabric free to move only a rough estimation could be made and it was decided to use half the area of a circle made by the span of the parafoil tracing the circumference, resulting in a surface area of $\approx 3m^2$. This is an attempt to model the slider pulling the ends of the parafoil together and the deflation of the parafoil cells. Further wind tunnel testing is recommended to find an accurate area and drag coefficient, although for the purpose of this preliminary sizing the drag on the payload has a greater impact. Furthermore it was assumed that the payload rests in an upright position.

6.2 Results

The aforementioned methodology was implemented at the return cruise condition of 1500ft altitude, 30ms^{-1} horizontal velocity, 0ms^{-1} vertical velocity and the associated masses and line lengths of the first and second stages. The resulting angle, with respect to the vertical, at the helicopter attachment point, the maximum tension and minimum required diameter are presented in Table 5.

Table 5: Summary of the results of the payload equilibrium during return cruise.

Stage	Angle (degrees)	Maximum tension (kN)	Minimum required diameter (mm)
First	35.4	18.9	7.2
Second	45.3	10.1	5.2

Schematics of the resulting slung line along with convergence plots of both models can be found in §B where it is shown that the algorithm converges for both stages. Furthermore, analysis from [22] confirmed that the grapple line length and cruise speed were acceptable as they did not cause instabilities in the helicopter.

7 Grapple Control Device

The purpose of the grapple control device is to act as a mechanism to fine tune the position of the grapple in order for maximum likelihood of capture by acting to damp any swinging motion of the grapple and maintaining directional stability. A novel control device is proposed aiming to increase the likelihood and safety of recovery.

Many configurations were considered to achieve the control of the longitudinal and vertical positions including mounting a quadcopter underneath the grapple and the addition of lifting surfaces. However these were discarded, due to unnecessary complexity and control implications, in favour of a simple rudder.

7.1 Configuration

Much like a vertical stabiliser on a fixed wing aircraft, the rudder has inherent directional stability, and the anticipated low frequency lateral oscillations can be countered by a control surface deflection. As a result, these control modes are not the pertinent design drivers for this preliminary design. Instead, it is the control of the longitudinal and vertical position of the grapple that is most important. The final configuration was decided to be a split drag rudder.

The split drag rudder acts like a conventional rudder in that the control surfaces can deflect left and right to generate a lateral force in both directions. However the control surface can simultaneously deflect in both directions to produce drag. This simple solution allows the control of longitudinal, lateral and vertical position of the grapple while only using one control mechanism. It can control the vertical position because, as a result of the grapple line, there is a coupling between the longitudinal and vertical positions. The rudder does not experience high aerodynamic loads and hence can be manufactured from a CFRP.

7.2 Control Surface Design

As previously discussed, the preliminary design driver for the rudder is to produce as large a range of drag as possible in order to offer the largest range of longitudinal and vertical positions as possible, while also offering the flexibility to provide lateral forces to kill any unsteady oscillations.

The two options considered were Traditional Drag Rudders (TDRs) and Morphing Drag Rudders (MDRs). MDRs are currently the subject of much investigation into the control of flying wing aircraft, like the B-2 Spirit. They can create a seamless change in camber and preliminary studies indicate up to a 30%

increase in C_d is achievable compared to TDRs [23]. However, the correct actuation mechanism must be carefully designed in order to satisfy the design requirements. Considering the increase in drag compared to a TDR the MDR was selected.

7.3 Control Surface Actuation

The design drivers for the actuation of the MDR were reliability, simplicity, control authority and actuation frequency. Given that the MDR was selected to outperform a TDR it must therefore be capable of consistently reaching the large deformations required without failure. Furthermore, in order to save on installation and maintenance costs the design must be simple as well to warrant a successful cost benefit analysis compared to a TDR. In addition, the surface must be able to deflect at the required frequency otherwise effective control will not be possible. While high frequency deflection will not be required, the frequency must be high enough to allow satisfactory control. Three configurations were considered.

The first was a Shape Memory Alloy system (SMA), by applying an electric current to a SMA the structure deflects. However, additional power is required to hold the deflection in place and very low actuation frequency, of the order of 1 Hz, limits the control authority of the device. Furthermore, preliminary designs indicate significant reliability issues [24]. For these reasons this design was rejected.

The second was a chordwise continuous camber technology presented in [23]. A pre-tensioned elastomeric skin is attached to a braced pin jointed framework supported by a beam spring. An actuator applies a bending moment to the beam spring causing a deflection. However, simpler solutions exist that reduce the high part count and mechanical complexity of the pin jointed mechanism. For this reason the design was rejected.

The selected design was a fish bone active camber (FBAC) mechanism [25]. Inspired by the flexible structure of a fish's spine, the pin jointed framework is replaced with a single, 3D printed ABS plastic spine. Dyneema tendons connected between a trailing edge cap and a pulley can be used to deform the morphing section. The pulley is connected to an actuator through the use of a worm gear, meaning that no additional power is required to hold a given deflection. A lightweight Elastomeric Composite Skin (EMC) is pretensioned so that the compression surface does not buckle under large deformations. Due to the EMC skin being capable of achieving strain rates of 100%, fatigue and failure of the skin is not an issue [27]. Figure 5 outlines the design of the mechanism.

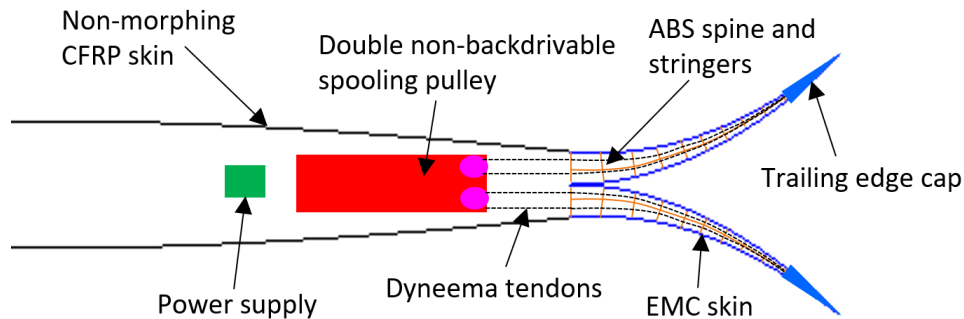


Figure 5: Schematic of the FBAC control surface actuation mechanism.

7.4 Sizing

A NACA 0012 aerofoil section of chord 1m, chosen to accommodate the various systems, was to be extruded for a given span. To reduce unwanted drag due to any sharp edges a rounded cowling was added onto the top and bottom surface of the rudder. The design point is the likely interception altitude of 10,000 ft, following the payload glide slope with a 35ms^{-1} horizontal velocity and a 15ms^{-1} decent velocity.

In order to find the optimal span for the rudder the methodology outlined in §6 was used to find the orientation of the rudder under the range of deflections permissible. Using a study at an equivalent Reynolds number, the maximum achievable C_d was assumed to be 0.6 at a deflection angle of 30 degrees, with the rear quarter of the section capable of morphing [23].

It was observed that, as expected, the larger the span the further back and the higher up the grapple could be taken to with full deflection. However, the further back the grapple was at zero deflection due to the increased wetted area, reducing the permissible range of positions. As a result, it was concluded that the optimal area was a function of the zero and full deflection drag coefficient and the mass of the grapple and rudder.

An iterative process was conducted with the aerodynamics and structures team to find the optimal surface area. CFD simulations were run to find the zero deflection C_d of the rudder and the results can be found in [28]. The masses of the required systems were estimated by scaling that of a prototype FBAC concept [26]. The mass breakdown of the optimal rudder can be found in Table 6.

Table 6: Summary of the mass components of the optimal rudder.

Component	Actuator	Power supply	CFRP structure	ABS spine	Tendon	EMC
Mass (kg)	0.45	0.73	3.0	1.75	0.01	0.5

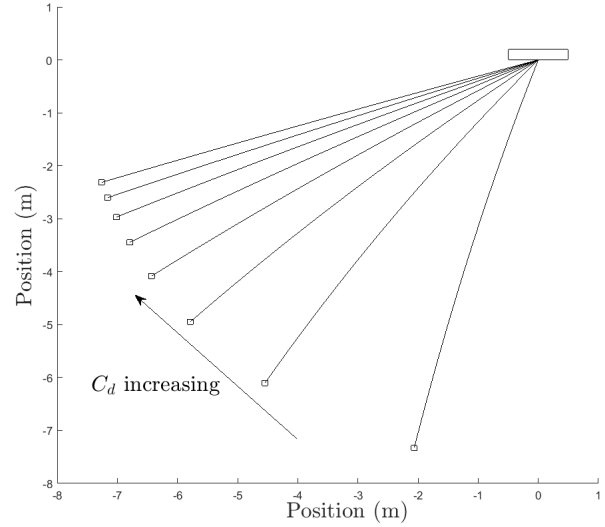
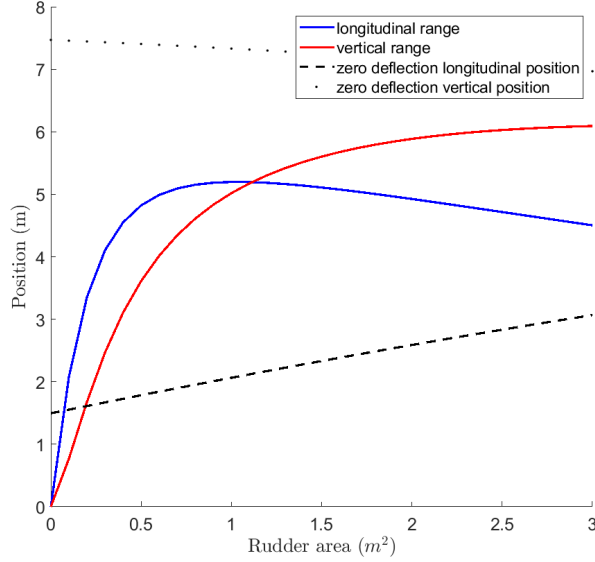
Figure 6a shows the variation of the longitudinal and vertical range of grapple positions with surface area, along with the zero deflection position of the grapple. It can be clearly seen that as the zero deflection position moves further back the range of longitudinal positions reduces, the optimal surface area was found to be 1.08m^2 as this allows the greatest variation of longitudinal position. The grapple can move 5.2m in the longitudinal direction and 5.0m in the vertical direction. Furthermore it allows a safe comprise of vertical positions, preventing the rudder impacting the aircraft.

Figure 11 in section §C shows a detailed schematic of the rudder and grapple integration as well as the cowling placement.

8 Simulation of Recapture Force

In order to determine if the aircraft is capable of responding, in a satisfactory manner, to the forces experienced during recapture, a model of the loading time history during the recapture was generated. More specifically, at the moment the parafoil is reefed by the action of the slider.

It was first considered to evaluate the loading history experienced during the deployment of the parafoil, which is well documented in the literature for slider reefed configurations [29], and reverse that history to find the force on the aircraft. However, the underlying physics of these models does not accurately represent the correct mechanisms involved in MAR. These models are driven by the imbalance of the friction in the slider system and the aerodynamic drag created by the deployment of the drogue and parafoil, which is a function of the payload velocity and altitude. On the contrary, during MAR the loading history is dependant on the relative displacement of the payload and the aircraft as the reefing action is purely displacement driven. To this end a two stage displacement driven model was implemented.



(a) Variation of grapple position with increasing surface area.

(b) Visualisation of the permissible range of positions of the grapple

Figure 6: Selection of the optimal surface area for maximum control authority and the resulting range of permissible longitudinal and vertical positions.

The first stage consists of a linear mapping of the relative vertical displacement between the payload and aircraft to the force transmitted onto the aircraft due to the collapse of the parafoil cells. The onset of the load is when the relative displacement is equal to the combined length of the grapple and the drogue lines. Physically this means those lines are in tension and any further increase in the relative displacement will move the slider up and the parafoil will begin reefing. The load stops when the displacement is further increased by the total length the slider must travel. It is assumed that the trajectory of the payload is unchanged during the loading which is a reasonable assumption given the relatively small time scales involved in the transient load. The stage 1 force can be expressed as

$$F_1(t) = F_{steady} \frac{d(t) - d_0}{l_{slider}} \quad (1)$$

where F_1 is the stage 1 force, F_{steady} the steady state force, d the relative displacement, d_0 the relative displacement at the onset of the force and l_{slider} the distance the slider must travel.

The second stage models the dynamic behaviour associated with the dissipation of inertial loads. In order to simulate the physical processes of the oscillation and damping of the dynamic response it was decided to model this as the response of a second order system and hence having a response defined by

$$F_2(t) = F_{steady} \left[1 - \frac{1}{\sqrt{1 - \zeta^2}} e^{-\zeta \omega_0 t} \sin \left(\omega_0 t \sqrt{1 - \zeta^2} + \cos^{-1}(\zeta) \right) \right] \quad (2)$$

where F_2 is the stage 2 force, ω_0 the natural frequency and ζ the damping ratio. Note that the first section of this response from $F = 0$ to $F = F_{steady}$ is already modelled by stage 1 and hence the response after this is taken.

Figure 7 shows the generation of the load history using flight test data from a recovery attempt. The values of ω_n and ζ are 50 and 0.4559 respectively resulting in a load factor of 1.2. Stage 1 lasts for 1.15s while the settling time for stage 2 is 0.4s which are reasonable time scales for this process. More information on the recovery simulations can be found in [30]

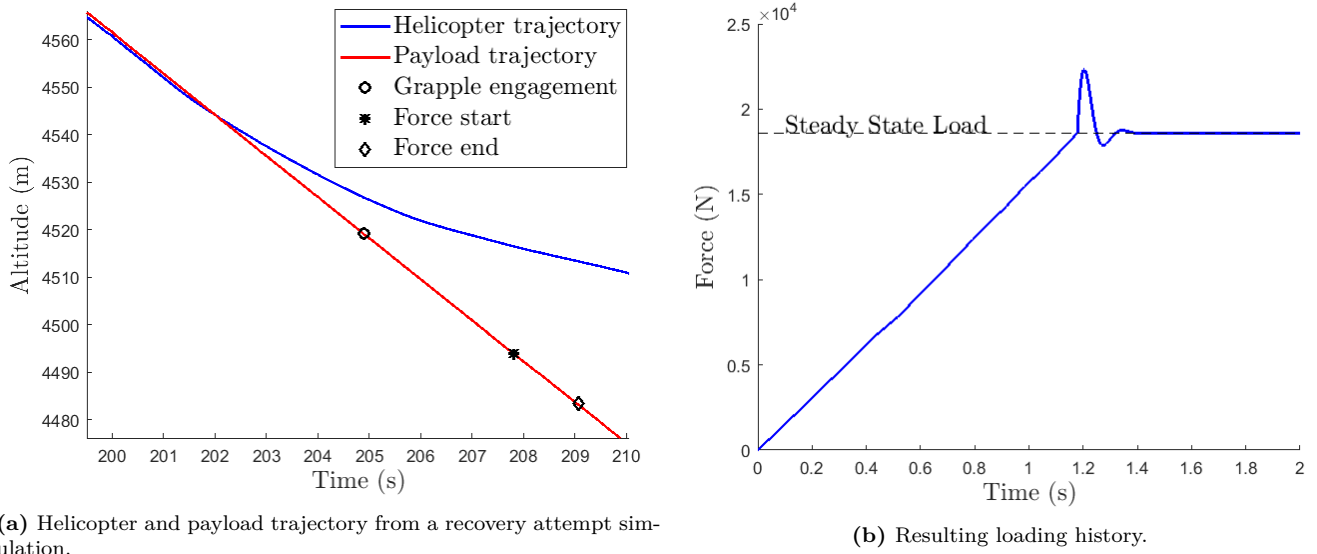


Figure 7: Flight test data of a recovery attempt and the resulting load history.

The advantage of this model is that by changing the values of ω_0 and ζ , one can quickly prescribe a set of expected loading histories of which to test, allowing for the determination of a range of acceptable and unacceptable responses of the aircraft. However, due to resolution issues in X-Plane, the flight test cannot accurately model the stage 2 response with values of $\omega_0 > 50$. Despite this, a pilot would not realistically be able to react to such a high frequency input meaning that the low resolution will not greatly impact the flight test. While the model serves as an effective tool for preliminary design, the complex loading history during recovery would be best modelled by real flight tests which would also analysis of the effect of the changing payload trajectory during the loading.

9 Conclusion

The preliminary design of the system architecture for the recapture of an orbital launch system has been designed. The most appropriate recovery method has been selected to ensure maximum likelihood of recovery and through the proposal of novel solutions, existing technology has been augmented. The safety is dramatically increased with the design of a remote piloted helicopter as the selected recover vehicle.

Preliminary sizing of the grapple hook was completed and a novel IR system was proposed to allow automatic tine closure, reducing the already high pilot work load during recapture. Further work must be done to investigate the effect of varying the length of the grapple times and the implementation of the IR system. Additionally, more accurate estimation of the aerodynamic characteristics of the hook must be done.

Initial sizing of the grapple line was done using existing data and stability analysis of the resulting system indicated the designed cruise velocities would not cause instabilities of the helicopter. However, the system dynamics are highly non-linear and extensive flight tests must be employed to validate those predictions.

A novel grapple control device was designed using a MDR for increased positional control allowing for an increased likelihood of recovery over conventional control devices. However, the validation of achievable C_d values must be completed to include 3D effects building on the results from [23].

The proposed 2 stage loading model was effective in assessing the aircraft's stability during recapture, however it does not account for the complex interactions and flight test validation is required.

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A Preliminary Weight and Power Sizing of Manned and Unmanned Tiltrotor Concepts

Table 7 summarises the results of the estimation of a manned and unmanned tiltrotor concept. The payload mass was 3000 kg and the time for hover and cruise was 40 minutes.

Table 7: Summary of the initial sizing of a manned and unmanned tiltrotor concept.

Configuration	Empty mass (kg)	Power (kW)	Cruise fuel mass (kg)	Hover fuel mass(kg)
Manned	5000	3600	420	1800
Unmanned	4500	3200	370	1600

B Payload Equilibrium Condition Results

Figure 8 shows the steady state orientation of the payload for the first and second stage. The freestream conditions are an altitude of 1500ft, forward velocity of 30 ms^{-1} and decent velocity of 0 ms^{-1} .

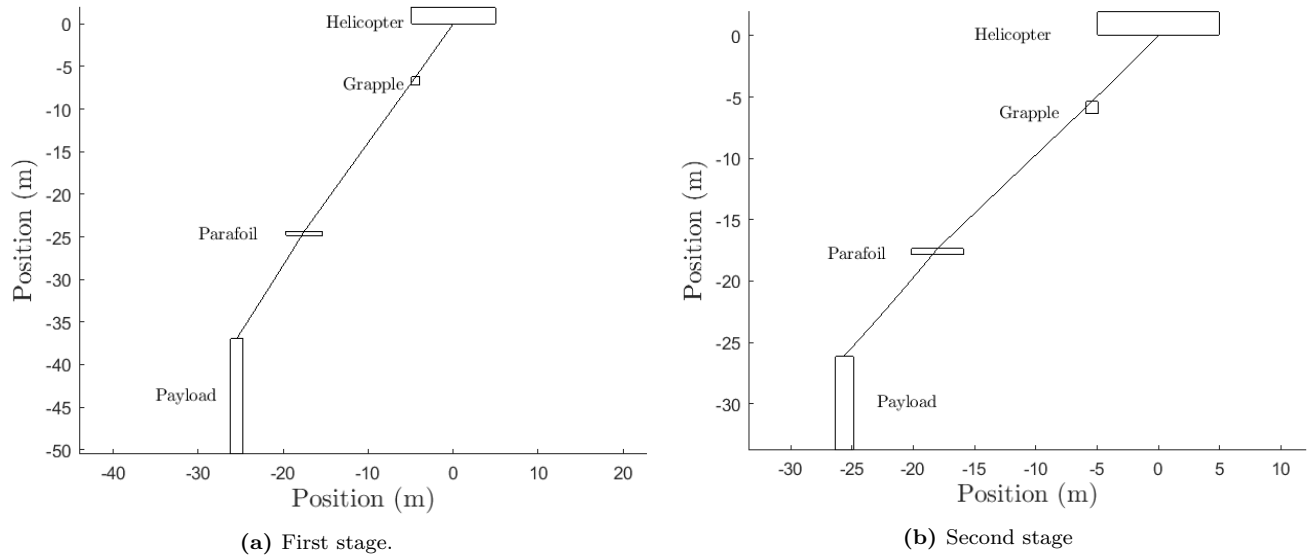


Figure 8: Schematic of the equilibrium state of the payload during return cruise.

Figures 10 and 11 show the convergence of the diameter and tension in the grapple line following each iteration. Note the first iteration is not included as arbitrary values were assigned to the tension and diameter for the first iteration. Both schemes converge as can be seen by the error between each iteration decreasing to a negligible value.

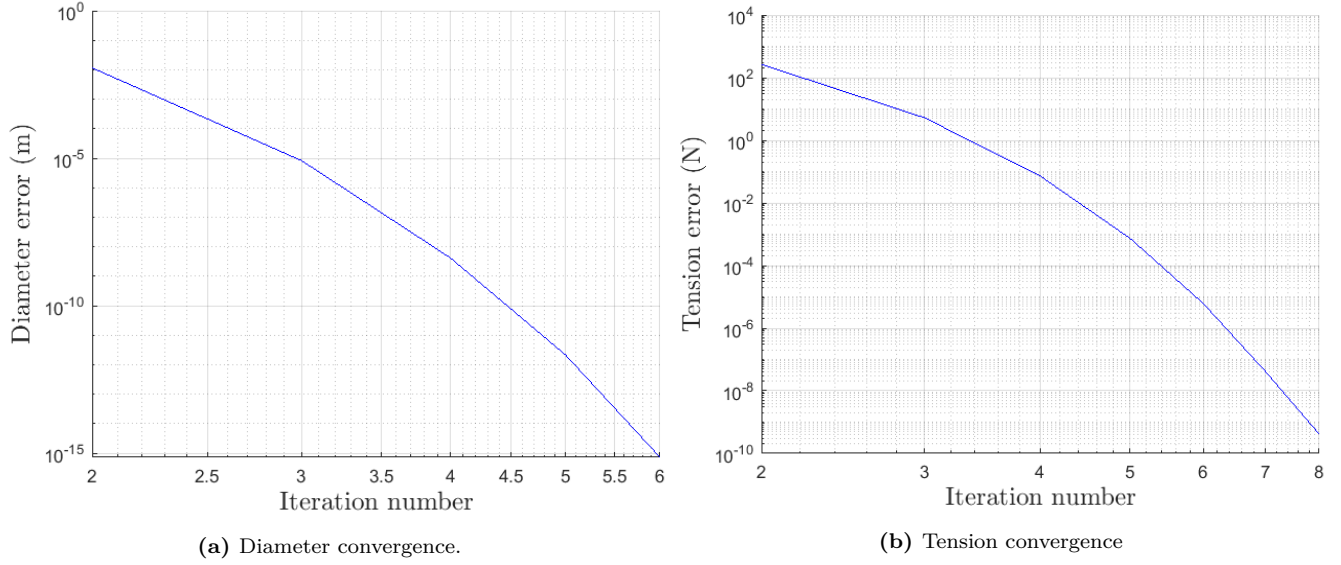


Figure 9: First stage equilibrium position convergence results.

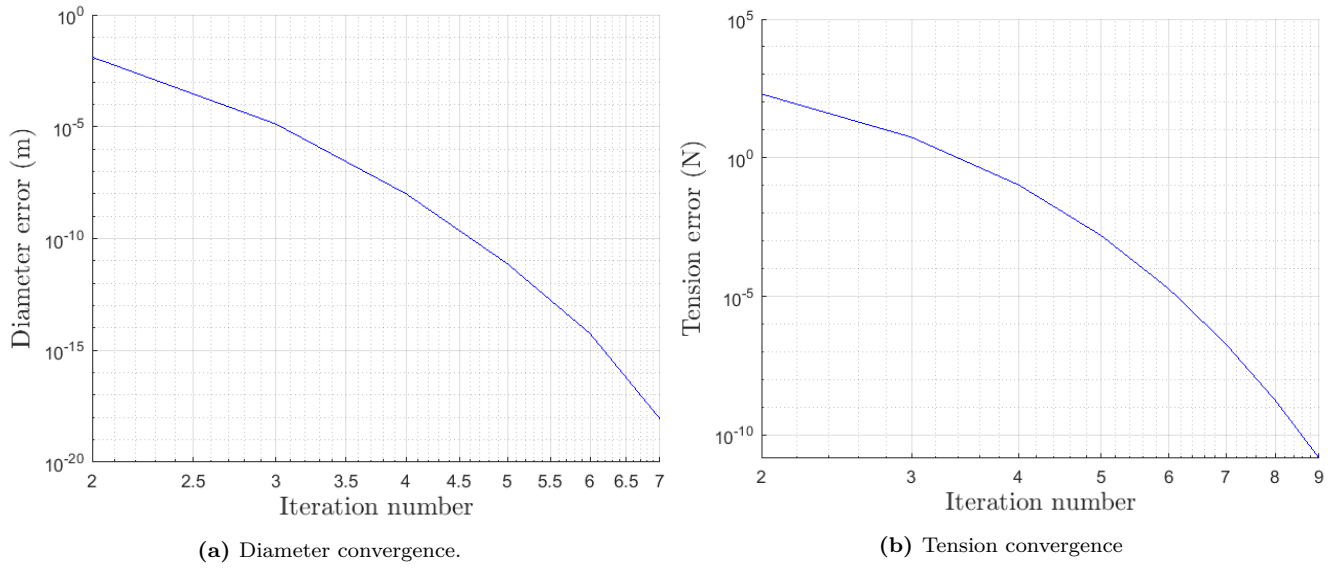


Figure 10: Second stage equilibrium position convergence results.

C Grapple Rudder Schematic

Figure 11 details the interior schematic of the grapple and grapple rudder. As shown in the figure the freestream is from left to right. Furthermore the cowling is a revolution of half a NACA 0012 section for reduced zero deflection drag.

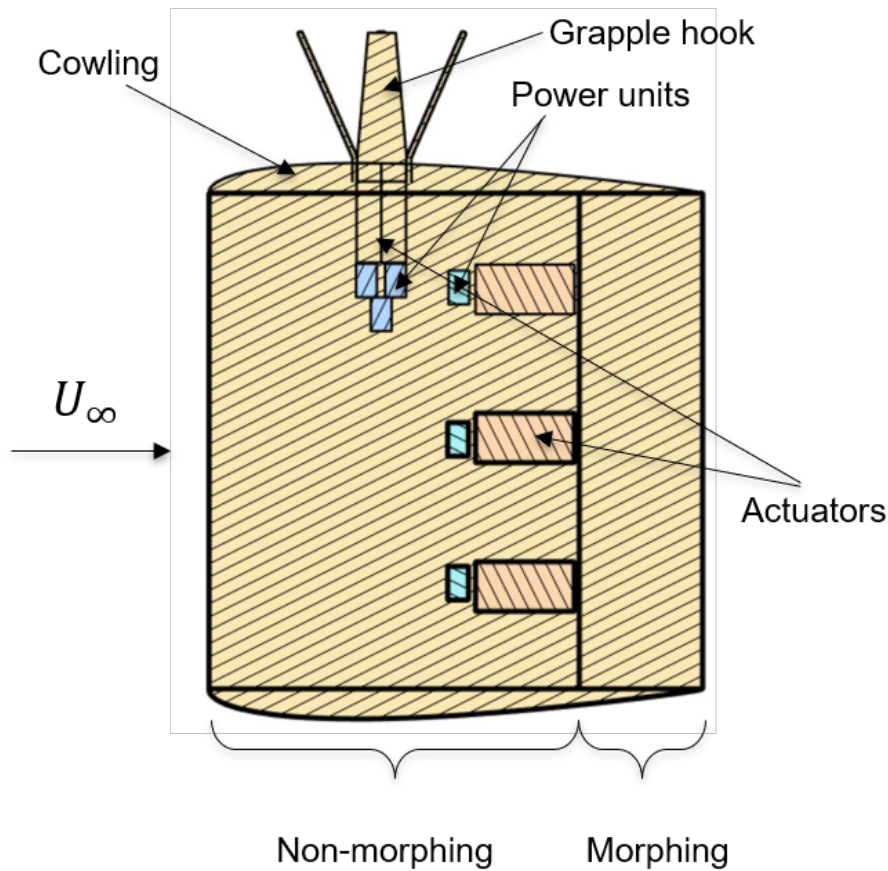


Figure 11: Side view schematic of the grapple and grapple rudder.

D Detailed Schematics of the Recovery Procedure

Figures 12 to 16 outline steps 1 through 5 of the recovery procedure. Note the front view is such that the payload is travelling out of the page, the side view is such that the payload is travelling from left to right and the red arrows indicate the direction of motion at any step. Furthermore, the grey objects represent the outline of the previous step to enhance the clarity of the procedure.

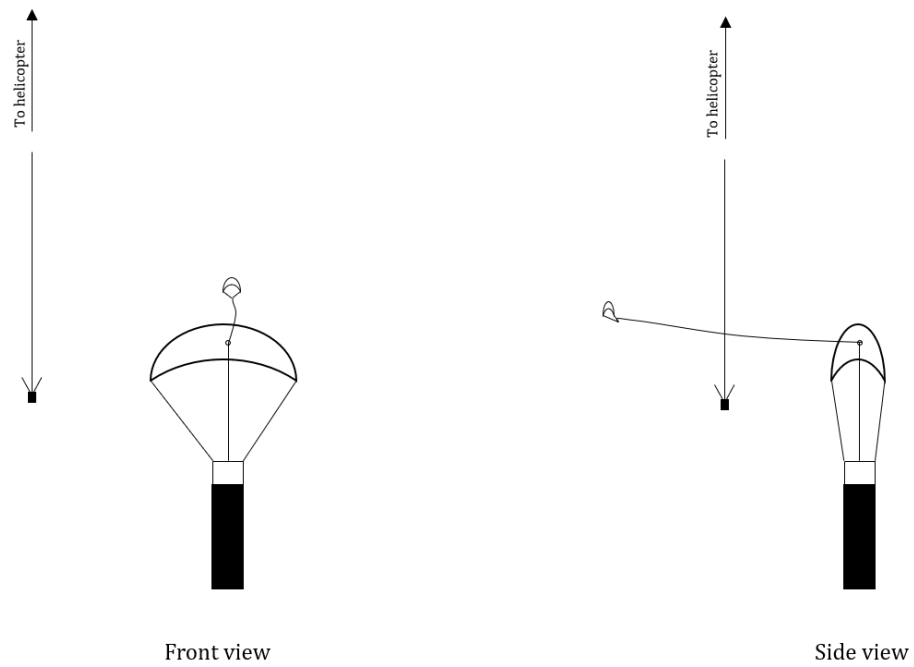


Figure 12: Recapture step 1.

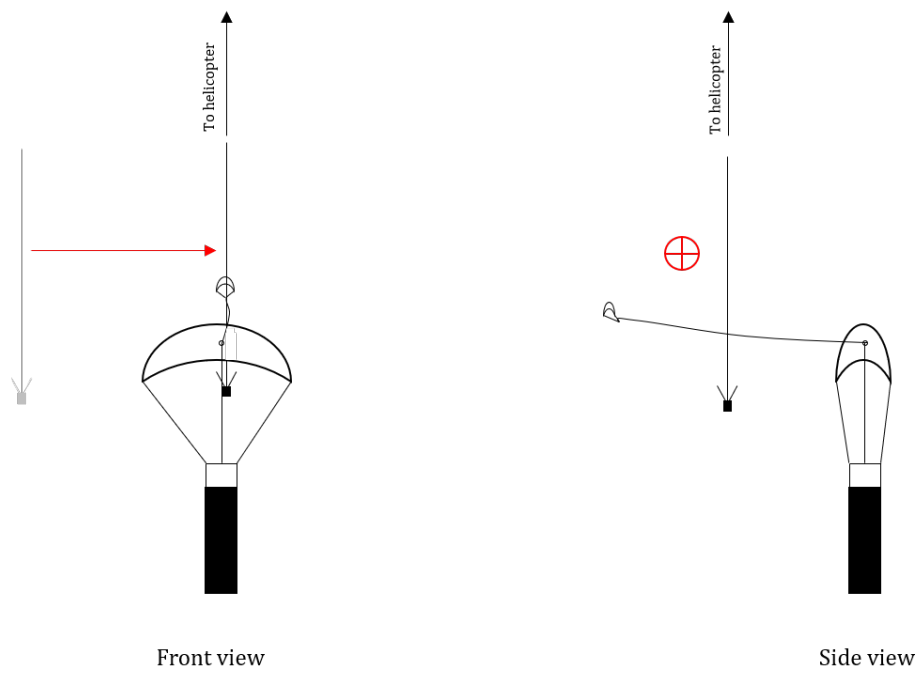


Figure 13: Recapture step 2.

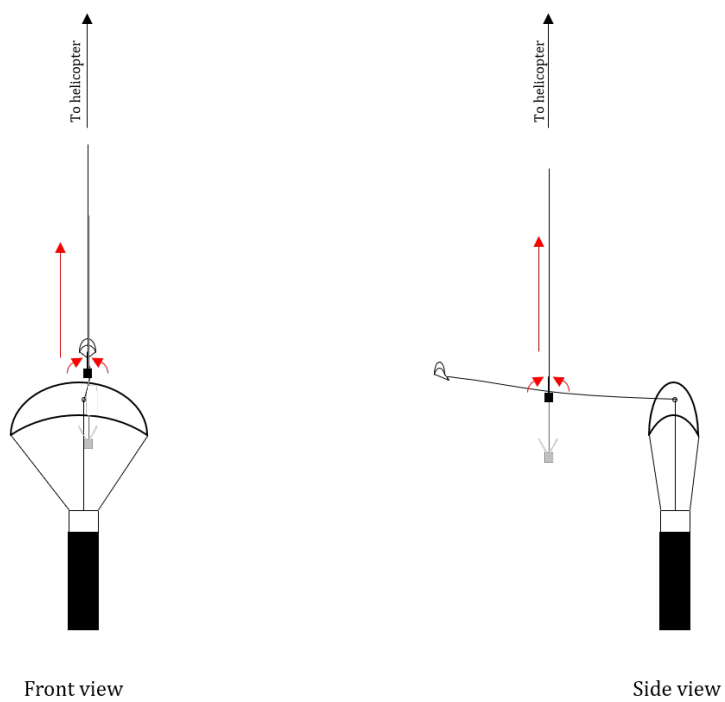


Figure 14: Recapture step 3.

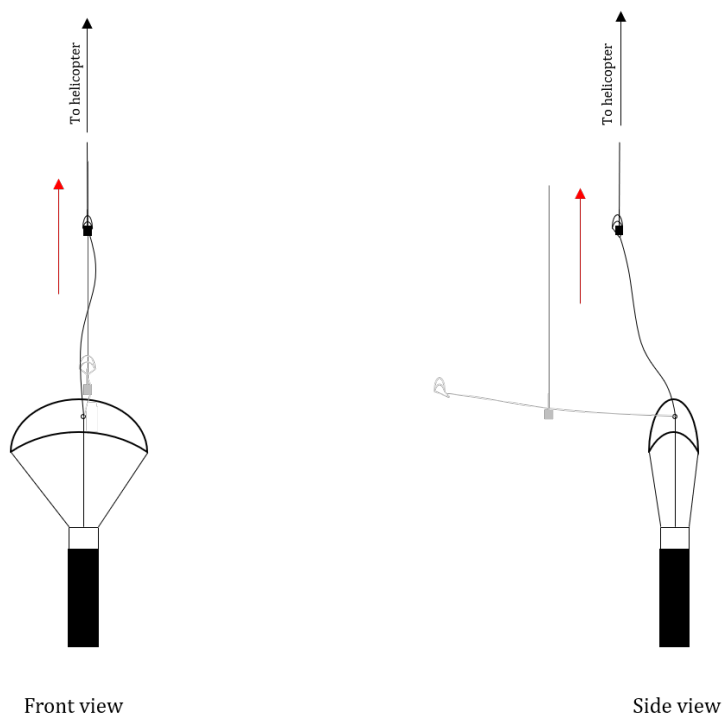


Figure 15: Recapture step 4.

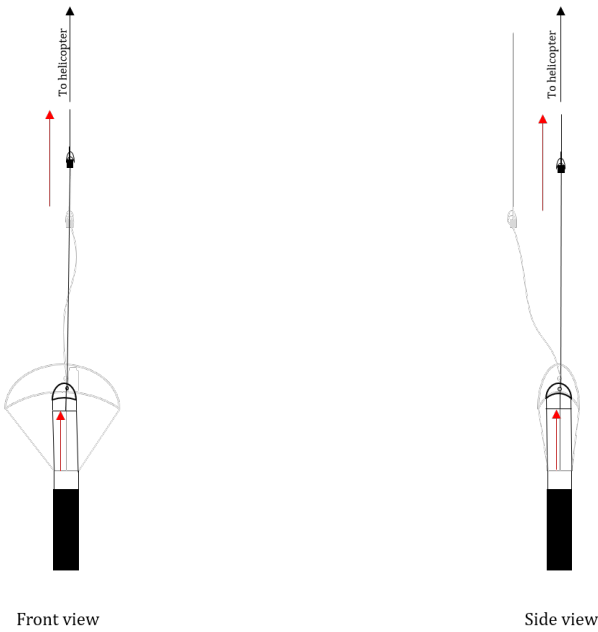


Figure 16: Recapture step 5.